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Impact of Fracturing Geometry on the Selection of Propped Fracturing, Acid Fracturing, and Matrix Acidizing: A Numerical Simulation Study

J. Deng, D. Stemberger, M. B. Smith, and D. R. Guerra, NSI Technologies, Tulsa, OK, USA; M. Y. Lezghem, S. Goyal, S. Al-Shammmary, and R. Chahine, Gulf Drilling & Maintenance Company, Al Ahmadi, Kuwait

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Abstract

This study evaluates the effectiveness of propped fracturing, acid fracturing, and matrix acidizing as oil and gas well stimulation techniques for carbonate reservoirs. There are several excellent studies on this topic in the literature with all concluding that as permeability of the formation decreases, propped fracturing becomes more favorable with longer fractures being more important. However, these studies uniformly implicitly assume that it is possible to create these longer propped fractures – but what if that is not the case?

The selection of the optimal stimulation method is influenced by factors such as reservoir properties, formation damage mechanisms, and geomechanical parameters. In the case of fracturing, fracture geometry significantly impacts the contact area between the fracture and the reservoir, with excessive fracture height growth (out of the pay zone) adversely affecting stimulation efficiency. For tight reservoirs, propped fracturing is usually preferable when fracture height is well constrained, although this confinement is not always achieved. Numerical simulations are utilized to examine how fracture geometry affects the optimal stimulation technique selection.

Introduction

Carbonate reservoirs, significant sources of hydrocarbon, are prevalent across regions such as the Middle East, the North Sea, North America, etc. A primary distinction between carbonate and sandstone reservoirs is the chemical reactivity of their minerals. Carbonates are generally soluble in most acids, whereas sandstones are more chemically stable. Therefore, the stimulation techniques for carbonate reservoirs include not only propped fracturing but also acid fracturing and matrix acidizing (without fracturing). The selection of the stimulation technique depends on reservoir petrophysics, mineral composition, geomechanical properties, and other operational considerations.

For a deep carbonate reservoir subject to high in-situ stresses, propped fracturing is generally more effective than acid fracturing, because the placement of proppants sustains the fracture conductivity. However, [Neumann et al. \(2012\)](#) have reported that acid fractures withstand high stresses in deep and hard limestones offshore Brazil. Later, [Oliveira et al. \(2014\)](#) continued to compare matrix acidizing and propped

fracturing. It is recommended that a set of criteria be developed to guide the selection of the stimulation technique for each candidate well. Suleimenova et al. (2016) performed laboratory experiments on core samples from deep and low permeability reservoir at a depth of 5,000-7,000 meters. Under that elevated closure stress condition, propped fractures retain greater conductivity compared to acid fractures. Jeon et al. (2016) studied key influencing factors from various reservoir types and proposed a guideline for the treatment selection. Bale et al. (2010) combined acid fracturing with proppant (CAPF) and suggested a revolutionary approach that takes advantages of both acid and proppant.

Of the numerous studies reviewing the topic of stimulation techniques, recent studies by Schwalbert et al. (2020 and 2024) compare proppant fracturing, acid fracturing, and matrix acidizing for carbonate reservoirs.

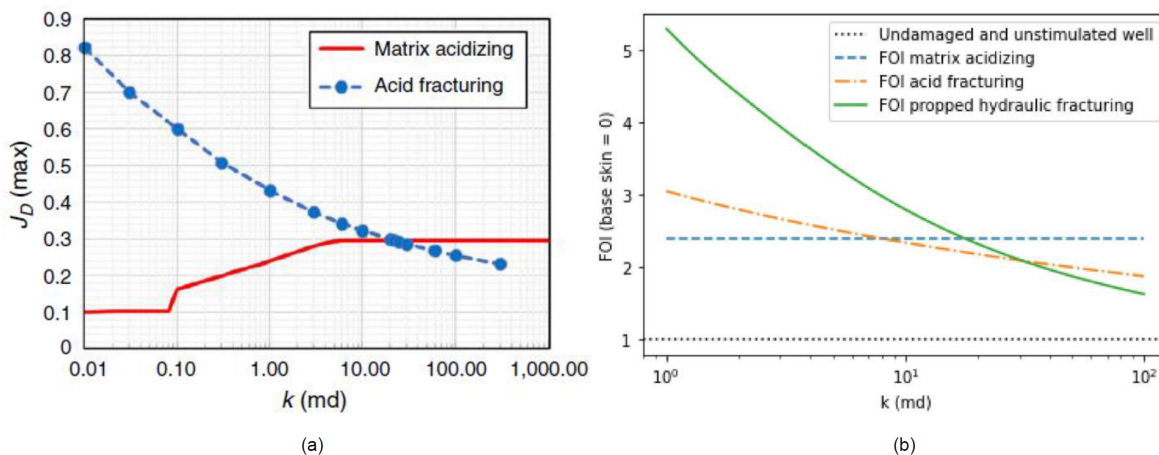


Figure 1—Comparison of different stimulation techniques for carbonate reservoirs: (a) Maximum productivity index for matrix acidizing and acid fracturing (Schwalbert et al. 2020); (b) Folds-of-Increase (FOI) for matrix acidizing, acid fracturing, and propped fracturing (Schwalbert et al. 2024).

In Fig. 1a, Schwalbert et al. (2020) showed that matrix acidizing has little productivity enhancement when formation permeability is below approximately 0.09 mD. From 0.09 mD to 0.5 mD, there is an increase in dimensionless productivity index, J_D , until a maximum of 0.3 is reached. Above these permeabilities, the J_D does not increase. Matrix acidizing is the preferred method of stimulation of carbonate reservoirs above formation permeability of approximately 10 mD. Below 10 mD, acid fracturing provides higher J_D than matrix treatments. Proppant fracturing as a stimulation technique was not studied.

Schwalbert et al. (2024) expanded their study to include proppant fracturing to the stimulation portfolio. In Fig. 1b, they calculated the Folds of Increase (FOI) to provide a graphic description of where each stimulation technique provided the maximum production increase. They found that proppant fracturing maximized FOI for every formation below permeability of approximately 15 mD. This then implies that for any formation with a permeability below 15 mD, there is only one stimulation technique available to maximize FOI. Nowhere in this discussion is economics nor the ability to physically create the fracture geometry required by each stimulation technique. In no case is acid fracturing the preferred case and in the absence of economics and derived fracture geometry is excluded as the best stimulation technique.

Comparative evaluations of the three stimulation techniques often assume effective fracture height containment; however, this assumption does not consistently reflect field conditions. In-situ stress profiles vary among formations and generally increase with depth. As a result, fractures may extend beyond the targeted pay zone and propagate into adjacent lower-stress intervals. This out-of-zone growth increases the likelihood of near-wellbore screen-out during propped fracturing and may yield limited stimulated fracture length within the productive interval. When propped fractures propagate downward, the settling of proppant can result in reduced conductivity within the pay zone. Therefore, fracture geometry and containment behavior must be integrated into the evaluation framework when comparing stimulation techniques.

Mahrer (1999) provides a comprehensive review of published studies on hydraulic fracture geometry, demonstrating that fracture configurations are more complex than conventionally anticipated. The fracture geometry depends on the native conditions of formation and treatment procedures. Numerous papers are published from the Hydraulic Fracture Test Site 2 (HFTS2). Ugueto et al. (2021), based on interpretation of HFTS2 data, report that hydraulic fractures tend to be predominantly planar and aligned with the direction of maximum horizontal stress. It exhibits strong fracture upward growth tendencies due to initial stress state and depletion in shallower formations. Wang et al. (2022) analyzed low-frequency distributed acoustic sensing (LF-DAS) data from a vertical monitoring well (HFTS2) and found excessive hydraulic fracture height growth on the order of 1,000-1,500 ft with upward extension significantly exceeding downward propagation. Weimann, et al. (2024) investigated the fracture height growth in Vaca Muerta by examining some variables, such as principal stress state, laminations, and lithological contrast.

Recent investigations into fracture height growth have primarily focused on unconventional reservoirs; however, excessive vertical fracture propagation can also occur in carbonate formations. In this study, a series of numerical simulations were conducted to evaluate the influence of fracture geometry on the selection of stimulation techniques in carbonate reservoirs. The analysis includes a comparison of initial post-treatment production performance over the first year, coupled with an economic assessment. The objective is to establish a framework for optimizing stimulation method selection based on fracture geometry.

Methodology

The numerical investigation encompasses the simulation of propped fracturing, acid fracturing, and matrix acidizing using commercially available software. This section delineates the computational methodology and outlines the governing equations implemented within the numerical models.

Propped Fracturing Models

During simulation the program solves a volume conservation equation for the fluid and proppant in the fracture which is continually changing length, height, and width (Smith et al., 2001).

$$\frac{\partial V_f}{\partial t} + \Delta_x q_x + \Delta_z q_z = q_p - q_l \quad (1)$$

V_f is the local fracture volume that is the hydraulic fracture width multiplied by the area of the fracture face. q_x and q_z are the clear fluid and proppant volumetric flow rates along the length and height of the fracture respectively. These flow rates are functions of the pressure gradient, the non-Newtonian viscosity, and the width. q_p is the pump rate; and q_l is the clear fluid loss rate at the fracture face. The equation is solved by finite-difference methods.

The volumetric flow rates are related to the pressure through the equations below.

$$2\left(\frac{4n'+2}{n'}\right)n'\frac{\mu}{w^2}f(V_{\text{prop}})q_x = -\frac{\partial p}{\partial x}A_x \quad (2)$$

$$2\left(\frac{4n'+2}{n'}\right)n'\frac{\mu}{w^2}f(V_{\text{prop}})q_z = -\frac{\partial p}{\partial z}A_z \quad (3)$$

$$\mu = K'\dot{\gamma}^{n'} \quad (4)$$

μ is the viscosity. n' is the flow behavior index. K' are the flow consistency index. f is a function related to the amount of proppant in the slurry. A_x and A_z are the fracture cross sectional areas in the fracture length and height directions, respectively.

The proppant particles are transported by the fluid. The force, F , on a particle in a flowing fluid uses the equation below.

$$F = C_D \frac{1}{2} \rho_f v_r^2 A_p \quad (5)$$

C_D is the drag coefficient. ρ_f is the fluid density. v_r is the relative velocity between the fluid and particle. A_p is the frontal area of the particle.

Normal one-dimensional Carter type fluid loss uses the simple equation. (Smith and Montgomery, 2015)

$$v_L \approx \frac{C_T}{\sqrt{t - \tau}} \quad (6)$$

v_L is the fluid loss velocity. C_T is the total fluid loss coefficient.

Acid Fracturing Models

For acid transport simulations, diffusion to the fracture face, reaction at the face, and leak-off into the formation are added to the basic fluid flow equation above. How quickly acid molecules can move from the interior of the fracture to the fracture face is a major factor in the amount of acid etching. This mass transport rate is governed by one of two phenomena: molecular diffusion (the most common controlling behavior) and convection. The equation for acid mass conservation is shown below.

$$\frac{\partial V_f C_a}{\partial t} + \Delta_x (C_a q_x) + \Delta_z (C_a q_z) + \Delta_y D A_y \frac{\partial C_a}{\partial y} + q_l C_{as} = q_p C_{ap} \quad (7)$$

V_f is the fracture volume. C_a is the bulk acid concentration. C_{as} is the surface acid concentration. C_{ap} is the acid injection concentration. q_x and q_z are the liquid lateral and vertical flow rates respectively. q_l is the liquid loss rate. q_p is the pump rate.

When acid molecules ($[H^+]$ ions) are consumed at the fracture face, a concentration gradient develops between the bulk and the surface. How quickly the ions can move from the interior of the fracture to the face is represented by the diffusion coefficient, D . The diffusion rate of the acid molecules can be expressed in terms of mass transfer coefficient. (Conway et al. 1999)

$$D \frac{\partial C_a}{\partial y} = \frac{Re D}{w} (C_a - C_{as}) \quad (8)$$

D is the mass transfer coefficient. Re is the Reynolds number. w is the total (hydraulic and acid etched) fracture width.

The reaction rate determines how rapid acid, in direct contact with the fracture face, can react to dissolve the calcite and etch the fracture face. The model assumes that the rate of diffusion of $[H^+]$ ions from the bulk to the face is in equilibrium with the rate of ion consumption at the surface. The surface reaction rate is (Buijse, et al., 2003)

$$R_a = k_0 \exp\left(-\frac{E_0}{RT}\right) \left\{ [H^+]_s^n - X^n \frac{(C_a - C_{as})^{2n}}{[H^+]_s^n} \right\} \quad (9)$$

R_a is the reaction rate. k_0 is the reaction rate constant. E_0 is the activation energy. R is the gas constant. T is the temperature. n is the reaction order. X is a constant.

Several acid etched conductivity models (Deng et al., 2012) are published. This study uses the classic Nierode and Kruk (1973) correlation.

$$k_f w = C_1 e^{-C_2 \sigma} \quad (10)$$

$$C_1 = 1.47 \times 10^7 w_a^{2.47}$$

$$C_2 = (13.9 - 1.3 \ln S_{RE}) \times 10^{-3}, 0 < S_{RE} < 20,000 \text{ psi}$$

$$C_2 = (3.8 - 0.28 \ln S_{RE}) \times 10^{-3}, 20,000 < S_{RE} < 500,000 \text{ psi}$$

$k_f w$ is the acid etched conductivity. σ is the closure stress. w_a is the acid etched width. S_{RE} is the rock embedment strength.

Matrix Acidizing Models

The matrix acidizing model simulates the acid penetration into reservoir by the transient flow using superposition time. (Nozaki, 2012)

$$-\frac{kh}{141.2\mu}(p_R - p_w) = \sum_{j=1}^n \Delta q_j [p_D(t_{D,n} - t_{D,j-1})] + q_n s_n \quad (11)$$

$$p_D = -\frac{1}{2} Ei\left(-\frac{1}{4t_D}\right), t_D = \frac{4.395 \times 10^{-6} kt}{\phi \mu c_t r_w^2}$$

k is the permeability. h is the pay thickness. p_R is the reservoir pressure. p_w is the wellbore pressure. q is the injection rate. s is the skin. ϕ is the porosity. c_t is the total compressibility. r_w is the wellbore radius. t is the time.

Several wormhole models (Furui et al. 2011) are published. This study uses the simple volumetric model (Economides et al. 1994) to predict the wormhole length.

$$r_{wh} = \sqrt{r_w^2 + \frac{V}{\pi \phi h P V_{bt}}} \quad (12)$$

r_{wh} is the wormhole radius. V is the injection volume. $P V_{bt}$ is the number of pore volumes of acid injected at the time of wormhole breakthrough at the end of the core.

Skin after acidizing treatment can be calculated from the wormhole length. (Nozaki and Hill, 2010)

When wormhole radius is greater than the damaged zone radius, $r_{wh} > r_d$,

$$s = -\ln \frac{r_{wh}}{r_w} \quad (13)$$

When wormhole radius is less than the damaged zone radius, $r_{wh} < r_d$,

$$s = \frac{k}{k_d} \ln \frac{r_d}{r_{wh}} - \ln \frac{r_d}{r_w} \quad (14)$$

k_d is the damaged permeability.

We will use all three above models to simulate fracture geometry that does not behave in an ideal fashion where the fracture is not constrained within the producing horizon. A number of cases will be reviewed to identify when non-ideal fracture geometry may exist and the subsequent determination for optimal treatment type.

Case Study

Input Parameters

For the case study several input parameters are required for the model to complete the simulations. The tables below list those model inputs. A generic carbonate (100% limestone) formation is chosen as the test formation with Table 1 showing the key input parameters. All simulations are based on vertical wellbore architecture.

Table 1—Input Parameters of Numerical Simulations

Input Parameters	Values
Reservoir Type	Oil
Lithology	100% Limestone
Reservoir Top	7,400 ft TVD
Reservoir Thickness	150 ft
Permeability	Example 1: 10 mD; Example 2: 0.01 mD
Porosity	15%
Water Saturation	20%
Reservoir Pressure Gradient	0.45 psi/ft
Closure Stress Gradient for Reservoir	0.73 psi/ft
Closure Stress Gradient for Stress Barrier	1 psi/ft
Young's Modulus	4 million psi
Poisson's Ratio	0.3
Fracture Toughness	2,000 psi/ $\sqrt{\text{in}}$
Total Fluid Loss Coefficient	Example 1: 0.002 ft/ $\sqrt{\text{min}}$; Example 2: 0.0005 ft/ $\sqrt{\text{min}}$
Drainage Area	160 acres
Bottomhole Flowing Pressure	500 psi

Pump Schedules

The pump schedule for propped fracturing will be the basis of all three stimulation techniques where we try to keep volumes pumped into the reservoir consistent. For propped fracturing we will use a 30# cross-linked gel (30 pounds of gel per 1000 gals of water) to create fracture width during pumping as well as for proppant carrying capacity. To ensure treating pressures are below our maximum threshold of 10,000 psi, a pump rate of 30 bpm has been chosen. The proppant chosen is 20/40 mesh natural sand. Stronger proppants like ceramics and intermediate strength proppants provide higher conductivity at higher closure stresses but are not considered in this study. The proppant concentration is ramped up in 1 ppa (pound per gallon added) increments to a maximum of 6 ppa. There is a need to get sufficient near wellbore connectivity, so a final concentration of 6 ppa should provide adequate conductivity in the near wellbore.

For the acid fracturing pump schedule, 28% gelled hydrochloric acid (HCl) was chosen to maximize reservoir dissolution and improve fracture conductivity. This acid was split into several stages alternating with 30# cross-linked gel. The cross-linked gel acts as a buffer between acid stages, providing greater viscosity, thus fracture width. One of the first to discuss this style of treatment were [Hendrickson and Cameron \(1969\)](#).

Table 2—Pump Schedule for Propped Fracturing

Clean Fluid Volume (gal)	Proppant Concentration (ppa)	Pump Rate (bpm)	Fluid Type	Proppant Type	Stage Proppant Mass (lbs)	Cumulative Pump Time (min)
20,000	0	30	30# Cross-Linked Gel	-	0	15.9
10,000	1	30	30# Cross-Linked Gel	20-40 Mesh Sand	10,000	24.2
10,000	2	30	30# Cross-Linked Gel	20-40 Mesh Sand	20,000	32.8
10,000	3	30	30# Cross-Linked Gel	20-40 Mesh Sand	30,000	41.8
10,000	4	30	30# Cross-Linked Gel	20-40 Mesh Sand	40,000	51.2
10,000	5	30	30# Cross-Linked Gel	20-40 Mesh Sand	50,000	60.9
10,000	6	30	30# Cross-Linked Gel	20-40 Mesh Sand	60,000	71.0
Total Clean Fluid Volume: 80,000 gal						
Total Proppant Mass: 210,000 lbs						

Table 3—Pump Schedule for Acid Fracturing

Fluid Volume (gal)	Pump Rate (bpm)	Fluid Type	Stage Pump Time (min)	Cumulative Pump Time (min)
5,000	30	30# Cross-Linked Gel	4.0	4.0
5,000	15	28 wt% Gelled HCl	7.9	11.9
5,000	30	30# Cross-Linked Gel	4.0	15.9
5,000	15	28 wt% Gelled HCl	7.9	23.8
5,000	30	30# Cross-Linked Gel	4.0	27.8
5,000	15	28 wt% Gelled HCl	7.9	35.7
5,000	30	30# Cross-Linked Gel	4.0	39.7
5,000	15	28 wt% Gelled HCl	7.9	47.6
5,000	30	30# Cross-Linked Gel	4.0	51.6
5,000	15	28 wt% Gelled HCl	7.9	59.5
5,000	30	30# Cross-Linked Gel	4.0	63.5
5,000	15	28 wt% Gelled HCl	7.9	71.4
5,000	30	30# Cross-Linked Gel	4.0	75.4
5,000	15	28 wt% Gelled HCl	7.9	83.3
5,000	30	30# Cross-Linked Gel	4.0	87.3
5,000	15	28 wt% Gelled HCl	7.9	95.2
Total Acid Volume: 40,000 gal				
Total Fluid Volume: 80,000 gal				

Matrix acidizing volumes were designed to mimic the acid fracturing acid volumes as well as using the same HCl acid strength. The treatment was simulated at a constant pump rate throughout the treatment.

Table 4—Pump Schedule for Matrix Acidizing

Fluid Volume (gal)	Pump Rate (bpm)	Fluid Type	Pump Time
40,000	1	28 wt% HCl	953
Total Acid Volume: 40,000 gal			

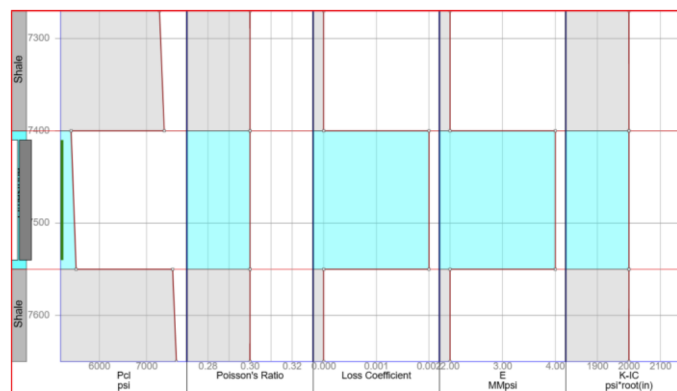
Geological Layering Scenarios

The following Table 5 shows the various sensitivities run during the study and is explained in detail following the table.

Table 5—Model Scenarios and Sensitivity Runs

Geological Scenarios		Example 1				Example 2			
		Permeability	Acid Frac	Prop Frac	Matrix Acid	Permeability	Acid Frac	Prop Frac	Matrix Acid
1	Contained Height	10 mD	✓	✓	✓	0.01 mD	✓	✓	×
2	Excessive Height Growth	10 mD	✓	✓	✓	0.01 mD	✓	✓	×
3	Limited Height Growth	10 mD	✓	✓	✓	0.01 mD	✓	✓	×

Geologic Scenario 1 is the case where the pay zone is sandwiched between two layers of impermeable rock layers with a stress difference greater than the net pressure that would be generated within the pay zone during stimulation. In this scenario it is assumed that the fracture will stay mostly contained and will thus generate fracture surface area almost exclusively within the pay zone.

**Figure 2—Geologic Layering for Scenario 1**

Geologic Scenario 2 is a case where the pay zone is adjacent to a thin higher stressed layer above it and a permeable non-pay zone above that. In this scenario, fracture height is not constrained within the pay, but a significant portion of the stimulation treatment could be lost to overlying zones.

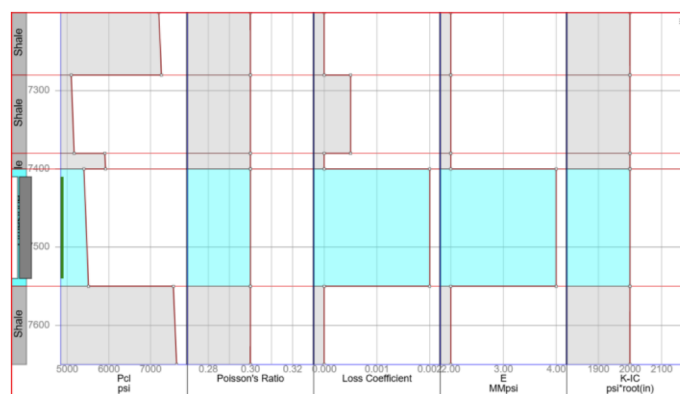


Figure 3—Geologic Layering for Scenario 2

Geologic Scenario 3 is a case where the pay zone is overlain with a thicker non-permeable zone with high stress which is in turn overlain by a lower stressed permeable zone. In this scenario there may be more fracture height generated with one type of treatment than the other.

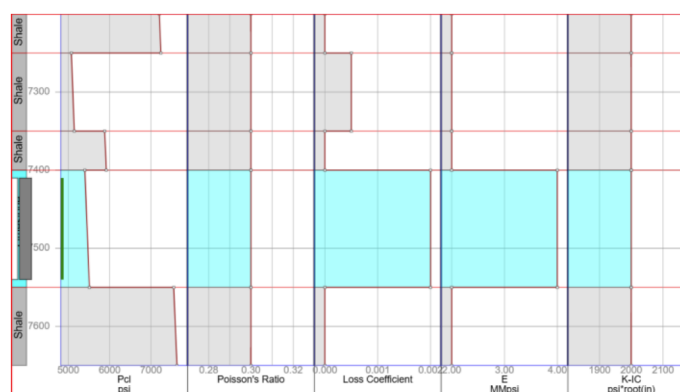


Figure 4—Geologic Layering for Scenario 3

The first set of simulations will focus on a reservoir permeability of 10 mD. All other reservoir and fracture parameters will remain the same for each of the stimulation cases. The initial skin factor will be assumed to be 0 for all cases. The three geological scenarios will be repeated with a formation permeability of 0.01 mD.

Example 1: Permeability of 10 mD

Matrix Acidizing. For the matrix acidizing simulation, we use an injection rate constrained model which holds injection rate constant throughout the treatment but ensures injection is under formation fracture pressure. Ideally, the treatment would start at low rates (always below fracture pressure) and as acid reaches the formation and skin is removed and stimulation takes effect, the injectivity would increase. As injectivity increases, the pump rate could be increased to rates above what would initially have been possible due to skin but still remain below fracture pressure. In Fig. 5 below we see the skin formation factor being reduced from zero at the beginning of the treatment to a final stimulation value of -3.2. The total treatment time was 953 minutes at a constant rate of 1 bpm.

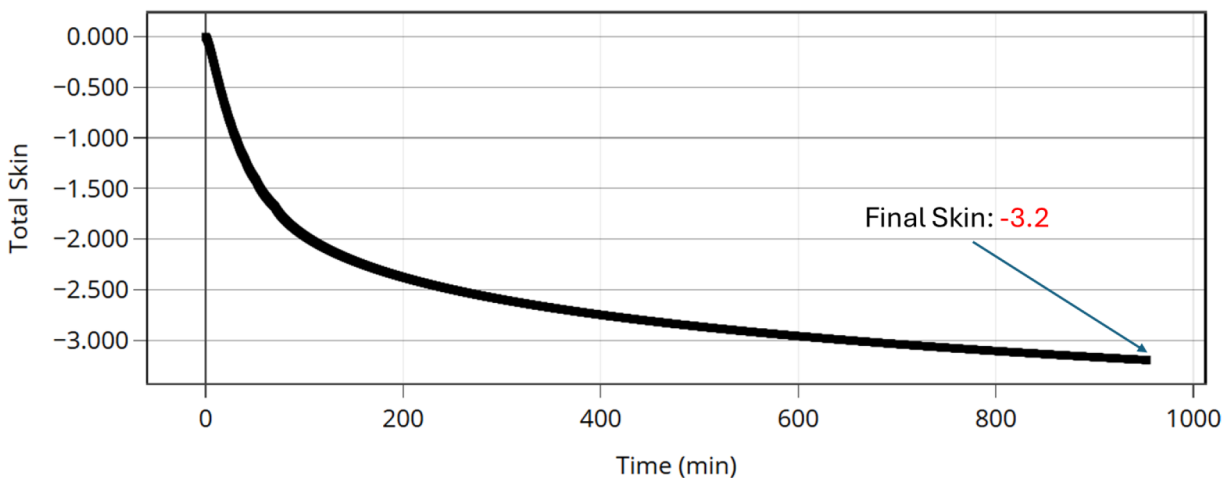


Figure 5—Matrix Acid Stimulation Post Stimulation Skin Improvement

Scenario 1: Contained Fracture Height. In Fig 6a below, we see a well contained propped hydraulic fracture with no proppant settling outside of the pay zone. Thus, almost all proppants remain in the pay zone contributing to production with practically no out-of-zone losses. The fracture grows with time until fluid leak-off leads to fracture tip extension cessation. The fracture half-length is about 350 ft with an average fracture conductivity of 1,755 mD-ft. In Fig 6b below, we see the acid fracture also remaining contained in-zone but with a lower average fracture conductivity of 2,030 mD-ft and extended conductive fracture half-length of 460 ft due to the gelled acid and cross-linked pad design.

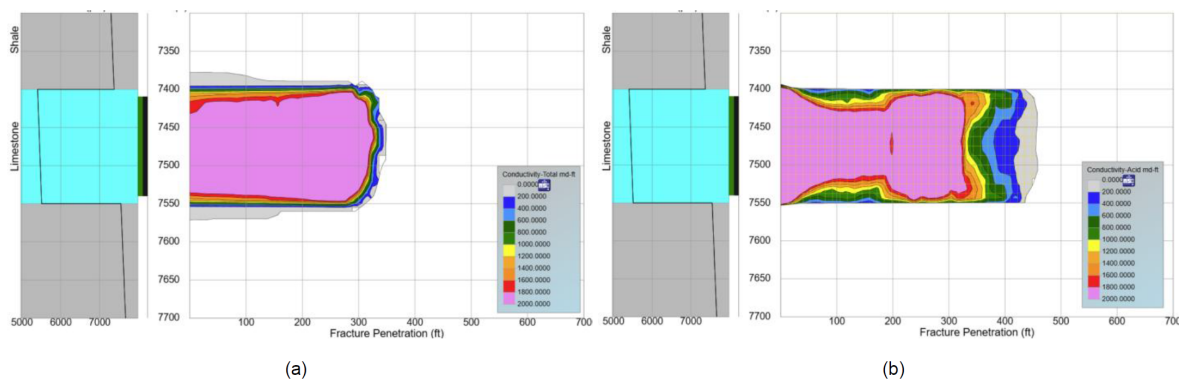


Figure 6—Comparison of fracture geometry and conductivity for scenario 1 of example 1: (a) Propped fracturing; (b) Acid fracturing.

The reservoir model used in this study is a 3-D numerical (finite difference, single phase) model. The results of the fracturing simulations were then run through the reservoir flow simulator to compare production with each of the techniques and are presented below in Fig. 7. From a flow standpoint, propped hydraulic fracturing is very similar to acid fracturing with matrix acidizing providing the least opportunity for maximizing productivity. Of note, the late-time production (1 year) shows all three types of stimulation produce at similar rates. The difference is in the cumulative production after one year where propped fracturing provides a FOI of 2.13, acid frac is 2.07 and matrix acid provides an FOI uplift of 1.57 compared to the unstimulated case. The key driver of resource capture is Net Present Value (NPV) or Return on Investment (ROI) which we will discuss in a later section of this paper.

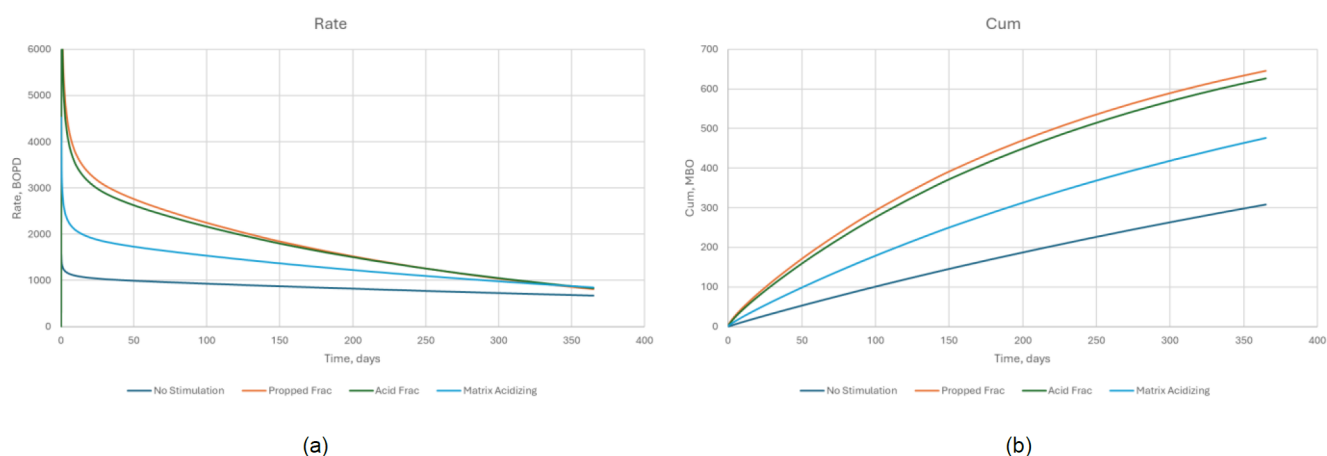


Figure 7—Comparison of post treatment production for scenario 1 of example 1: (a) Oil production rate; (b) Cumulative production.

Scenario 2: Excessive Fracture Height Growth. In this scenario, a low stress zone is situated immediately above the pay zone, separated by a thin barrier. In Fig. 8a, we see the results of the propped fracturing treatment where we break through the thin barrier and up into a lower stressed zone. Much of the fracture fluid and proppant is pumped into this lower stress zone, leaving less stimulation in the pay zone. In Fig. 8b, acid also breaks through to the upper zone, but since it is not acid soluble, no etching or long-term fracture width is created in this lower stressed zone. Much of the acid stimulation fluid is lost to the upper lower stressed zone, leaving the pay zone inadequately stimulated.

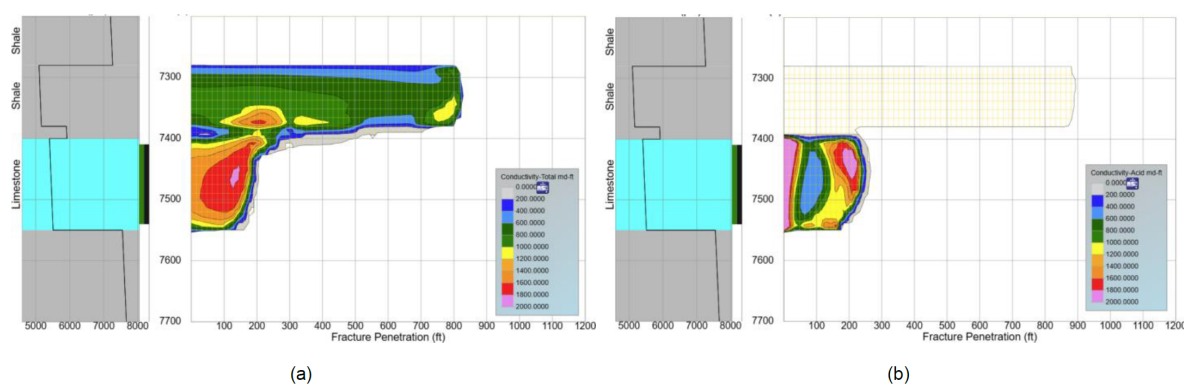


Figure 8—Comparison of fracture geometry and conductivity for scenario 2 of example 1: (a) Propped fracturing; (b) Acid fracturing.

The results of the fracturing simulations were then run through the reservoir flow simulator to compare production with each of the techniques and are presented below in Fig. 9 below. We again see that from a pure flow standpoint, propped hydraulic fracturing is by a narrow margin, the preferred stimulation technique and provides the best early time resource capture. Acid fracturing is, however, a close second. Matrix acidizing where 100% of the fluid is assumed to remain in the pay zone to create matrix wormhole conductivity. Of note, late-time production shows all three types of stimulation produce at similar rates. The less reservoir stimulation length leads to less cumulative production at one year (600 MBO versus 650 MBO in Scenario 1 for prop fracturing). In terms of cumulative production propped fracturing provides a one-year FOI uplift of 1.97, acid frac FOI uplift of 1.90 and matrix acidizing FOI uplift of 1.57.

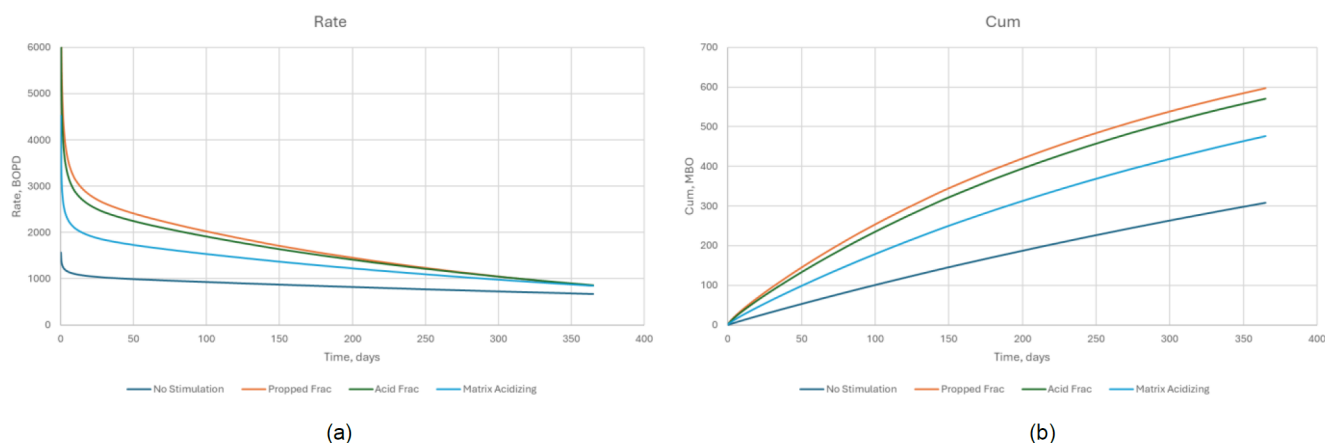


Figure 9—Comparison of post treatment production for scenario 1 of example 1: (a) Oil production rate; (b) Cumulative production.

Scenario 3: Limited Fracture Height Growth. In this scenario, there is a stress barrier that is not large enough to contain the propped fracture treatment (Fig. 10a) but does keep the acid fracture (Fig. 10b) contained. The reason for the propped fracture creating more fracture height is the high fluid viscosity and higher net pressure which drives the fracture upwards. Late-time breakthrough when net pressure exceeds breakdown pressure of the thick barrier allows some proppant to escape upwards, but much of the proppant gravity settles within the zone of interest. Acid fracturing does not create as much net pressure and remains contained. In this scenario, some of the propped fracture is lost to the upper lower stressed zone resulting in a shorter fracture half length. The acid fracture on the other hand remains constrained so all stimulation fluid (acid) is spent in the pay zone, thus rendering a stimulated fracture geometry like Scenario 1. There may be a critical net pressure that one can stay below in the prop fracture treatment, but this was not tested for this paper and may be a subject to cover in subsequent studies.

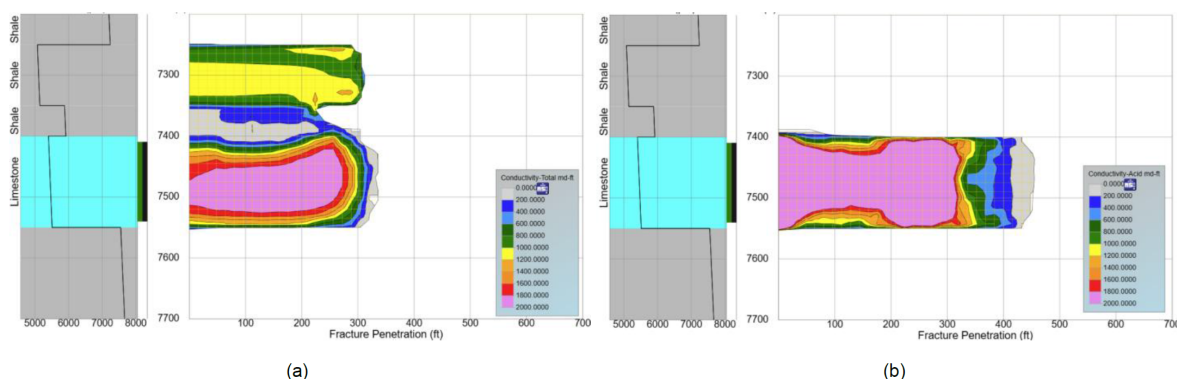


Figure 10—Comparison of fracture geometry and conductivity for scenario 3 of example 1: (a) Propped fracturing; (b) Acid fracturing.

Running the fracturing simulations results through the reservoir simulator provides the plots found in Fig. 11 below. As in all three cases, propped and acid fracturing provide similar production rates and cumulative production uplift with matrix acid a distant third. FOI increases here for propped and acid fracturing are 2.05 and for matrix acid, the same in all three cases at 1.57.

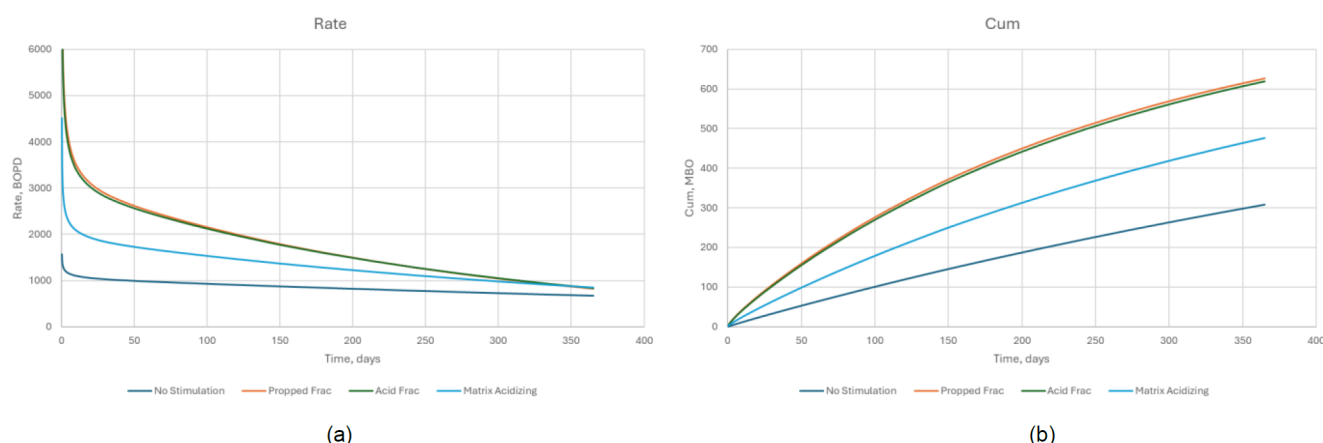


Figure 11—Comparison of post treatment production for scenario 3 of example 1: (a) Oil production rate; (b) Cumulative production.

In Fig. 12 below, we combined all three geological scenarios for proppant fracturing into one plot, Fig. 12a, and in Fig. 12b, combined acid fracturing and matrix acidizing. We see that the confined fracture scenario provides the best productivity in high permeability cases, where all treatments remain in the pay zone. Excessive height growth leads to an 8%-9% reduction in cumulative production after one year for both acid and prop fracturing. Matrix acidizing provides the least stimulation impact and is more or less immune from geomechanic differences in bounding layers. The key to optimizing the stimulation technique of choice will be to look at NPV or ROI.

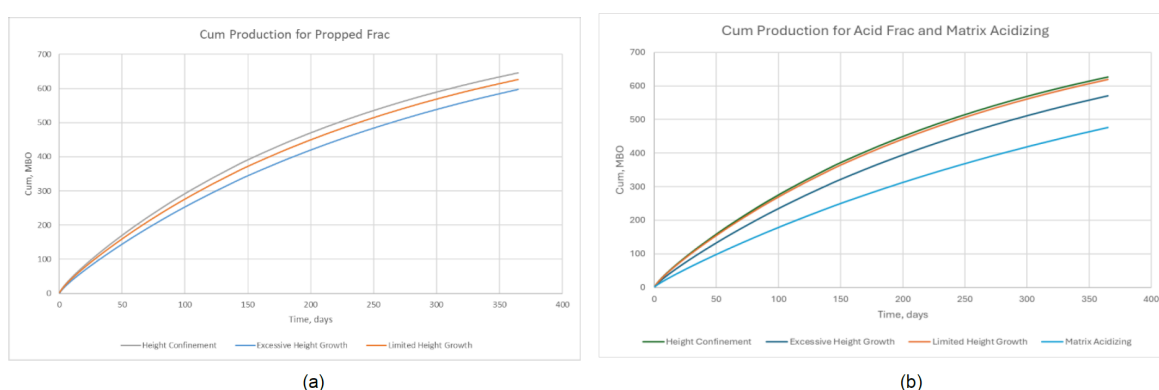


Figure 12—Cumulative production from different fracture geometry in example 1: (a) Cumulative production for propped fracturing; (b) Cumulative production for acid fracturing and matrix acidizing.

Example 2: Permeability of 0.01 mD

In this section we will look at the same three geologic scenarios, but this time decrease reservoir permeability and fluid leak-off as shown in Table 1 above. In tight reservoirs (typically considered unconventional), they do not produce at economic rates without fracturing stimulation. Furthermore, a purely matrix acid treatment is not typically warranted due to low injectivity rates and poor overall stimulation effectiveness. Schwalbert et al. (2020) showed that matrix acidizing is not effective at permeabilities below 0.1 mD. No matrix simulations were run due to long injection times that were not realistic. Instead, only the propped fracturing and acid fracturing treatment simulations were run in this example.

Scenario 1: Contained Fracture Height. In Fig. 13 below, we see that with low permeability and low fluid loss, fracture half lengths in both cases are more extensive. The fracture half-length of each technique exceeds 800 ft. Hydraulic fracture conductivity diminishes drastically with length (Fig. 13a) whereas acid fracturing conductivity is high throughout the fracture face.

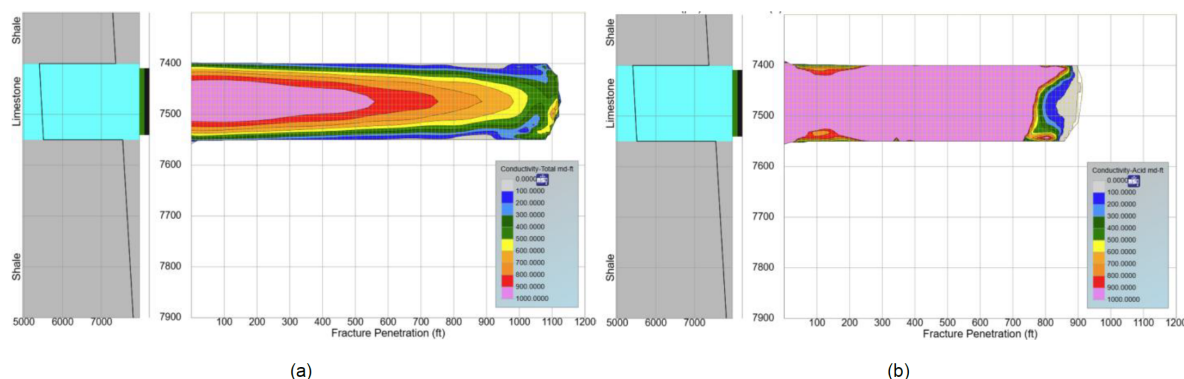


Figure 13—Comparison of fracture geometry and conductivity for scenario 1 of example 2: (a) Propped fracturing; (b) Acid fracturing.

Putting the fracture results into the reservoir flow simulator yields the production rate and cumulative production in Fig. 14a and 14b respectively. In the unstimulated case, we see basically zero production. With propped and acid fracturing, both have similar early-time production, however; acid frac conductivity diminished more with time than propped fracturing. Thus, propped fracturing has a slight edge in late-time production and cumulative total after one year. Fig 14a, shows proppant fracturing with a cumulative production increase after one year of 15.6% (37 MBO versus 32 MBO).

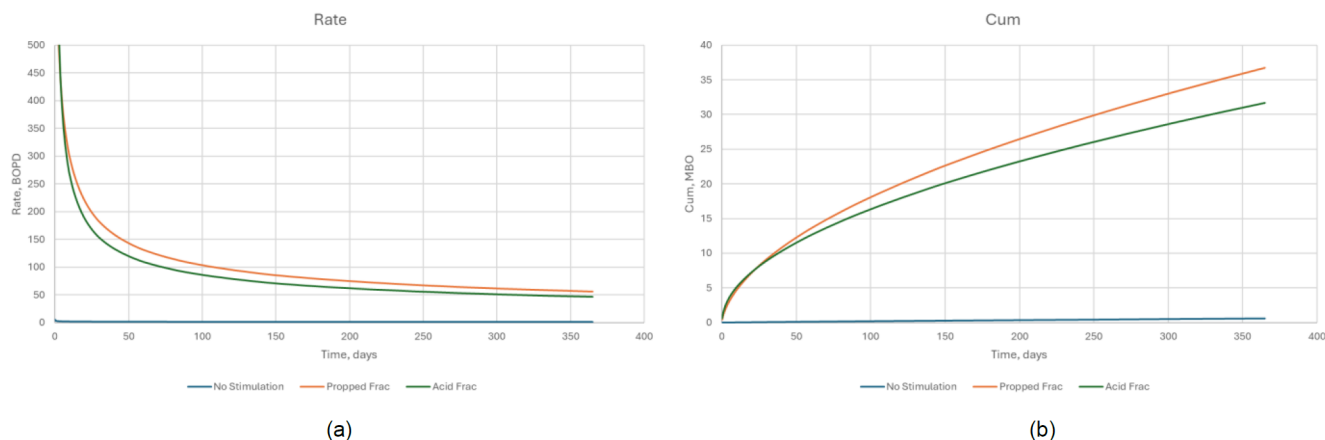


Figure 14—Comparison of post treatment production for scenario 1 of example 2: (a) Oil production rate; (b) Cumulative production.

Scenario 2: Excessive Fracture Height Growth. In Fig. 15a, we have excessive height growth out of the pay zone, we see significant loss of stimulation in the zone of interest and upward growth with proppant settling above the high stress zone and losing connectivity with the pay zone. The fracture half-length has decreased by 47% from the contained case (600 ft versus 1,130 ft). Acid fracturing in Fig. 15b also suffers from stimulation fluid loss to the low stress area above which leaves a shorter fracture half-length than anticipated (350 ft versus 910 ft). Comparing the propped to acid fracture half-lengths, propped frac has the edge 600 ft versus 350 ft (for acid) while the conductivity of the fractures is very similar.

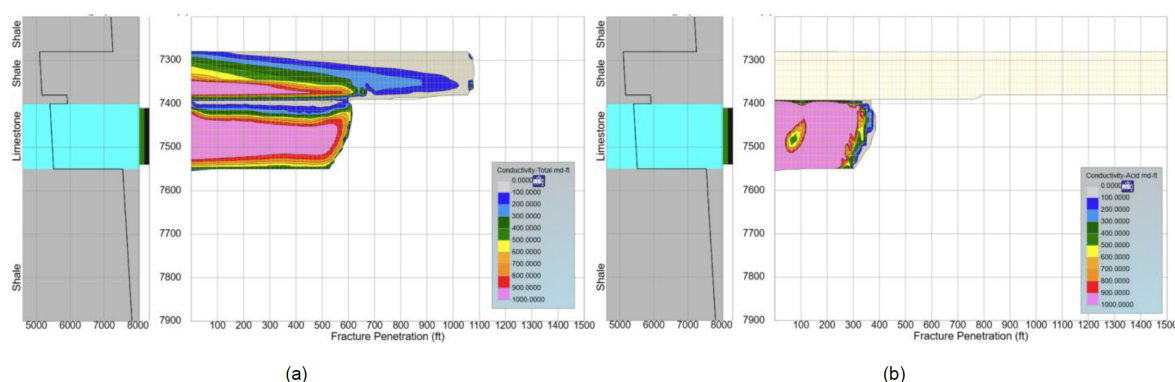


Figure 15—Comparison of fracture geometry and conductivity for scenario 2 of example 2: (a) Propped fracturing; (b) Acid fracturing.

Running the fracturing model results in the reservoir flow simulator provides the production rate and cumulative oil production as per Fig. 16a and 16b below, respectively. There is a similar production in early-time flow rate between propped and acid fracturing but 1 year cum production favors proppant fracturing by 56% (22.3 MBO versus 14.3 MBO for acid)

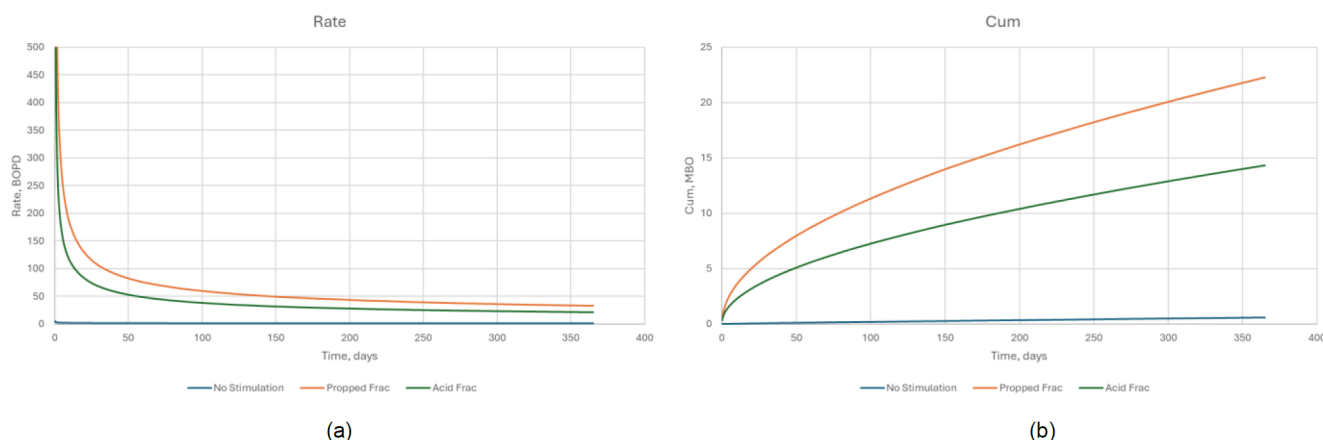


Figure 16—Comparison of post treatment production for scenario 2 of example 2: (a) Oil production rate; (b) Cumulative production.

Scenario 3: Limited Fracture Height Growth.. In our final simulation we look at Scenario 3 which has a thicker barrier. As with the unconstrained fracture height growth (Scenario 2) the propped fracture breaks through this barrier once the net pressure exceeds the breakdown pressure of the barrier and allows much of the fracturing treatment to break up into the lower stress area as per Fig. 17a. This results in a reduced fracture length of approximately 29% (800 ft versus 1,130 ft) from the constrained fracture Scenario 1. The acid fracture, Fig. 17b, is again unaffected as the net pressure never builds high enough to break through the barrier. This scenario then is the same as the constrained scenario in both 10 mD and 0.01 mD cases in that fracture half-length and conductivity are the same with no loss of stimulation treatment out of zone.

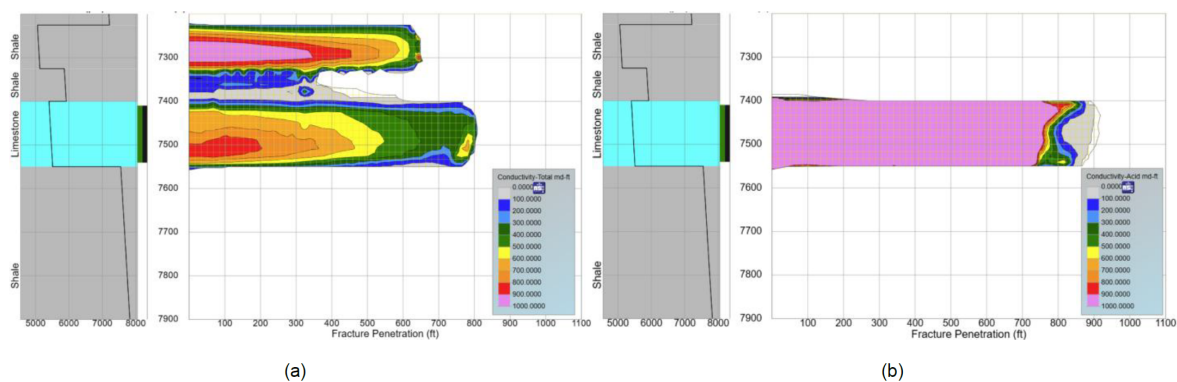


Figure 17—Comparison of fracture geometry and conductivity for scenario 3 of example 2: (a) Propped fracturing; (b) Acid fracturing.

Comparing productivity between acid and propped fracturing, production rate and cumulative production are presented in Fig. 18a and 18b respectively. We now see where acid fracturing now provides better initial and cumulative oil production. The loss of proppant to the upper zone reduces fracture half-length significantly to where the acid fracture now has a higher surface producing area than the propped fracture. The acid fracture provides a 15% uplift on FOI versus proppant fracturing (31.5 MBO versus 27.3 MBO)

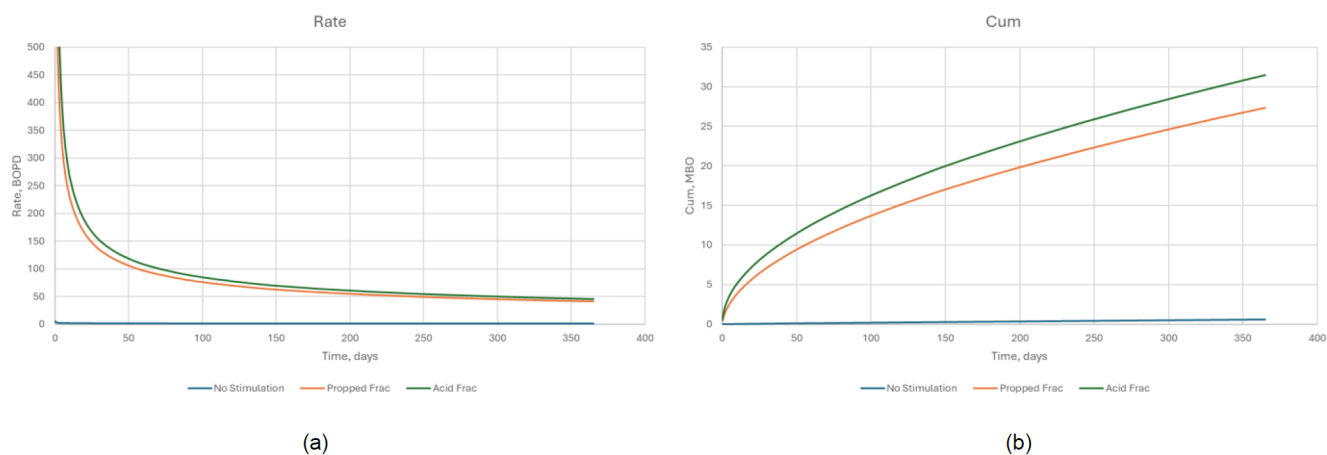


Figure 18—Comparison of post treatment production for scenario 3 of example 2: (a) Oil production rate; (b) Cumulative production.

Comparing cumulative production for all three scenarios in the low permeability (0.01 mD) case in Fig. 19, results in a major loss in productivity when the fracture grows out of zone. Propped fracturing has higher net pressures during fracturing and has the potential to have more height growth than acid fracturing. If the treatment can stay in zone, propped fracturing provides better stimulation and higher production than acid fracturing. This is not always the case, and acid fracturing may stay better contained than propped fracturing due to lower net pressure generation. A fully calibrated 3-D fracture geometry model is encouraged to be able to determine at which point (net pressure in the fracture exceeds barrier containment) fracture height confinement is lost, and treatment effectiveness is reduced. Fluid rheology and leak-off parameters are critical in developing a well calibrated simulation model.

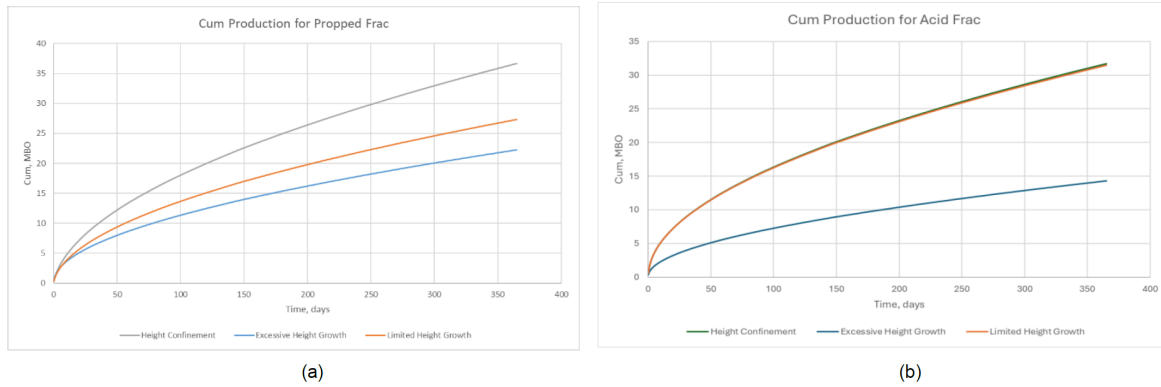


Figure 19—Cumulative production from different fracture geometry in example 2: (a) Cumulative production for propped fracturing; (b) Cumulative production for acid fracturing.

Economics Discussion

An attempt was made to use general Middle East pricing for fluids, equipment and services that were provided by a local vendor. There are too many variables to develop definitive pricing, instead a generic price was determined based on the volumes of fluid, proppant and horsepower required to deliver each style of treatment. Nothing else is factored into these costs like tree-saver requirements, flowback equipment, disposal, screen-outs, clean-outs, and numerous other concerns with each treatment style. The intent is to look at the big picture and help differentiate the production potential via Net Present Values (NPV) and Return On Investment (ROI) calculations.

NPV is defined as the difference between the present value of cash inflows and the present value of cash outflows over a period of time. We have simplified this to only include the cost of the stimulation and the uplift in production from the stimulation over the period of one year.

$$NPV = \frac{\text{cash flow}(\text{revenue})}{(1+i)^t} - \text{initial investment} \quad (15)$$

Cash flow (revenue) is simply one-year total production (bbls) multiplied by prevailing oil price (\$/bbl). i is interest rate. t is time periods (1 in our case for one year). Initial investment is the cost of stimulation. For comparison, we normalized NPV with propped fracturing set at 100%.

ROI measures the profitability of an investment relative to its cost. ROI is a ratio that compares the gain or loss generated from an investment relative to its cost. It is expressed as a percentage and is calculated by subtracting the initial investment from the final value (NPV) of the investment, dividing the result by the initial investment (cost of the treatment), and multiplying by 100. Rather than showing a percentage we normalized ROI with proppant fracturing set at 100%.

For our economic analysis we calculated ROI as follows:

$$ROI = \frac{NPV - \text{Initial Cost}}{\text{Initial Cost}} \quad (16)$$

Table 6 below shows the assumptions used in the calculations of NPV and ROI.

In the following tables we will present the economic return for all scenarios of each stimulation technique for each permeability example, namely 10 mD and 0.01 mD.

Table 6—Inputs for Economic Calculation

Economic Parameters	Values
Oil Price	\$70/bbl
Interest Rate	10%
Time Period	1 year
Propped Fracturing Treatment Cost	\$190,000
Acid Fracturing Treatment Cost	\$165,000
Matrix Acidizing Treatment Cost	\$110,000

As presented in [Table 7](#), for a reservoir permeability of 10 mD, propped hydraulic fracturing yields the highest (in green) NPV, whereas matrix acidizing results in the lowest (in red) NPV. Despite its comparatively lower economic return, matrix acidizing exhibits the highest normalized ROI due to its lower treatment cost. The ROI values for propped fracturing and acid fracturing are comparable under these conditions. In Scenarios 1, 2, and 3, propped fracturing demonstrates slightly better NPV performance relative to acid fracturing and superior to matrix acidizing. In terms of ROI, matrix acidizing provides better initial return on investment due to the relatively low cost of stimulation.

Table 7—Economics for Example 1 (10 mD)

	Frac Half Length ft	Avg. Frac Conductivity mD-ft	One-Year Production M-bbl	Value MMS	NPV MMS	Normalized NPV %	Normalized ROI %
Scenario 1: Contained Fracture Height							
Propped Frac	350	1755	646	45.2	40.9	100%	100%
Acid Frac	460	2030	627	43.9	39.7	97%	112%
Matrix Acidizing	-	-	476	33.3	30.2	74%	128%
Scenario 2: Excessive Height Growth							
Propped Frac	200	1500	597	41.8	37.8	100%	100%
Acid Frac	250	~1000	570	39.9	36.1	96%	110%
Matrix Acidizing	-	-	476	33.3	30.2	80%	138%
Scenario 3: Limited Height Growth							
Propped Frac	300	1300	627	43.9	39.7	100%	100%
Acid Frac	450	2020	618	43.3	39.2	99%	114%
Matrix Acidizing	-	-	476	33.3	30.2	76%	132%

Matrix acidizing was excluded from the analysis at permeabilities below 0.01 mD and is therefore not considered in the subsequent evaluation. As indicated in [Table 8](#), at a permeability of 0.01 mD, propped hydraulic fracturing results in a higher (in green) NPV and ROI than acid fracturing for Scenarios 1 and 2. In Scenario 3, acid fracturing outperforms propped fracturing in both NPV and ROI. This enhanced performance is attributed to more fracture height growth in the propped case, while the acid fracture remains confined within the pay zone and penetrates more laterally into the reservoir. The increased reservoir contact area is particularly advantageous in enhancing hydrocarbon production from unconventional formations.

Table 8—Economics for Example 2 (0.01 mD)

	Frac Half Length ft	Avg. Frac Conductivity mD-ft	One-Year Production M-bbl	Value MMS	NPV MMS	Normalized NPV %	Normalized ROI %
Scenario 1: Contained Fracture Height							
Propped Frac	1130	636	36.7	2.57	2.14	100%	100%
Acid Frac	910	3600	31.7	2.22	1.85	86%	99%
Scenario 2: Excessive Height Growth							
Propped Frac	600	>1000	22.3	1.56	1.23	100%	100%
Acid Frac	350	>1000	14.3	1.00	0.75	61%	64%
Scenario 3: Limited Height Growth							
Propped Frac	800	~500	27.3	1.91	1.55	100%	100%
Acid Frac	910	3450	31.5	2.20	1.84	119%	142%

Conclusions

After modeling thousands of fractures with a 3-D hydraulic fracture simulator, we noticed that many fracs do not remain contained within the pay zone. We had often wondered if there was any correlation between stimulation techniques and how the fracture or matrix acid contacted the pay and potential non-pay intervals and how does losing stimulation treatment to non-pay zones affect productivity and NPV. In the simulations we have determined that propped hydraulic fracturing always produces the highest NPV except in the case of when there is limited height growth with acid fracturing – but the propped fracture grows out of zone by a larger amount; thus, leaving more of the acid treatment contained within the fracture leading to a longer fracture with higher fracture conductivity than the propped fracture.

When looking at matrix acidizing, it never provides the highest NPV but in the high perm case it always provides the best ROI. This is due to its relatively low cost and the improvement in productivity in a high permeability case and moderate productivity (FOI) enhancement. Depending on how a company looks at its internal economics, one treatment may favor another from a simple cash flow perspective.

Using a fully calibrated 3-D fracture simulator, the engineer can run an unlimited number of scenarios for their particular reservoir and bounding layer scenarios which will allow better understanding of which of the three stimulation techniques would maximize productivity and optimize economics.

Future studies could test rock properties (such as Young's modulus, Poisson's ratio, embedment strength, pump rate, fluid viscosities etc.) to determine the relationship of these properties to stimulation effects and optimal stimulation design.

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